

Corollary 1. Suppose $\phi \in (0, 1)$. Then, in the limit (4) and $\lambda \rightarrow 0^+$, the test error with respect to the true data distribution P_1 , of the classical linear model $\hat{f}_{CL}$ defined in (3) is given by

$$E_{test}(\hat{f}_{CL}) \simeq \sigma^2 \phi / (1 - \phi) + (p_2^2 + p_2 p_1 \phi / (1 - \phi)) c^2. \quad (11)$$

Moreover, for fixed $c > 0$ and small $\phi \in (0, 1)$, it holds that

$$E_{test}(\hat{f}_{CL}) \simeq \sigma^2 d / n + p_2^2 c^2 + O(\phi^2). \quad (12)$$

In particular, if $c^2 = \Omega(1)$, i.e. bounded away from zero (corresponding to low-quality synthetic data), then $E_{test}(\hat{f}_{CL}) = \Omega(p_2^2 c^2)$: the scaling law plateaus unless $p_2 \rightarrow 0^+$, i.e. unless all but a vanishing proportion of synthetic data is discarded from the training dataset.

The result is empirically illustrated in Figure 3, where we see that even a small amount of low-quality synthetic data is enough to cause model collapse, whereby the test error of the model deviates from a perfect diagonal (ideal scaling law, corresponding to $p_2 = 0$, i.e. training on real data only).

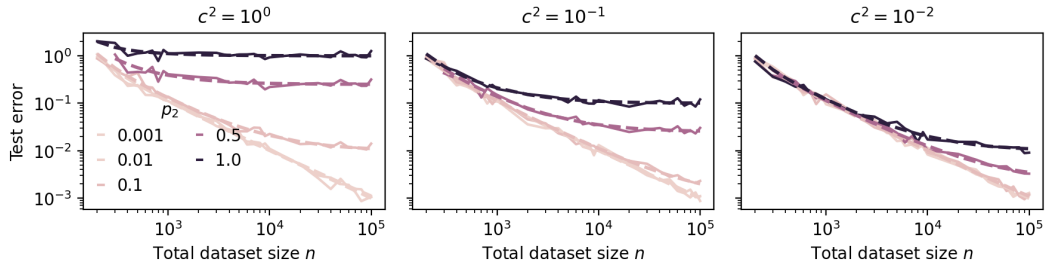


Figure 3: Strong model collapse in classical linear model (empirical confirmation of Corollary 1). The training dataset comprises of $n = n_1 + n_2$ samples from a mixture of $n_2 = p_2 n$ synthetic samples and $n_1 = n - n_2$ real samples. The real samples are from the same distribution as the real / true samples of the training dataset, while the synthetic samples are from a distribution with the same covariance structure and label noise level $\sigma = 1$, but an incorrect labelling function (epistemic error). The quality of the synthetic data is controlled by the scalar c , with $c \rightarrow 0$ corresponding to synthetic data of perfect quality (higher values correspond to lower quality synthetic data). Solid curves correspond to experiments, and broken curves correspond to our theoretical predictions of Corollary 1; notice the perfect match. We see that even a small amount of low-quality synthetic data is enough to cause model collapse, whereby the test error of the model deviates from a perfect diagonal (ideal scaling law, corresponding to $p_2 = 0$, i.e. training on real data only).

Remark 2. Corollary 1 can be extended to the non-isotropic case, but the statement is much longer and is thus omitted here.

3.2 Random Projections Model

We now turn to the more challenging setting of the random projections model $\hat{f}_{RP}$ given in (5). As mentioned before (cf. Section 2.2), such models are considered in our work as a simplification of the high-dimensional dynamics of actual neural networks, which still allows us to capture the effect of model size in the model collapse phenomenon.

We will need the following scalars which capture the high-dimensional statistics of the model $\hat{f}_{RP}$.

Definition 2. Let (e, τ) be the unique positive solution to the following fixed-point equations

$$1/e = 1 + \psi \tau \bar{\text{tr}} \Sigma K^{-1}, \quad 1/\tau = 1 + \bar{\text{tr}} K_0 K^{-1}, \quad \text{with } K_0 := e \Sigma, \quad K := \gamma \tau K_0 + \lambda I_d, \quad (13)$$

where $\bar{\text{tr}} A := \text{tr} A / d$ is the normalized trace operator. Also define (u, ω) by the equations

$$u = \psi e^2 \bar{\text{tr}} \Sigma_1 (\gamma \tau^2 L' + \omega I_d) K^{-2}, \quad \omega = \tau^2 \bar{\text{tr}} (\gamma \omega K_0^2 + \lambda^2 L') K^{-2}, \quad \text{with } L' := (1 + u) \Sigma. \quad (14)$$

Finally, define the following auxiliary scalars $\theta := \lambda / (\gamma \tau e) > 0$ and $\omega' := \omega / (\gamma \tau^2) > 0$.

As usual, we denote $\sigma^2 := p_1 \sigma_1^2 + p_2 \sigma_2^2$. Also, for any $p \in [0, 1]$ define a $d \times d$ positive definite matrix $T(\theta; p) := p \Sigma + \theta I_d$, and $T(\theta) := T(\theta; p)|_{p=1} = \Sigma + \theta I_d$.

The following result is a nontrivial extension of Theorem 1 to the case of random projections.

Theorem 2. In the proportionate scaling limit (6), the test error w.r.t the true data distribution P_1 , of the random projections model $\hat{f}_{RP}$ defined in (5) is given by $E_{test}(\hat{f}_{RP}) \simeq \bar{E} + \zeta$, with

$$\begin{aligned}\bar{E} &\simeq B + V, \text{ where } B = (1 + u)\theta^2 \text{tr} \Gamma \Sigma T(\theta)^{-2} + \omega' \text{tr} \Gamma \Sigma^2 T(\theta)^{-2}, \\ V &= (\text{tr} \Sigma^2 T(\theta)^{-2} + (\omega' - \theta u) \text{tr} \Sigma T(\theta)^{-2}) \sigma^2 / e, \\ \zeta &= p_2^2(1 + p_1 u) \text{tr} \Delta \Sigma^3 T(\theta)^{-2} + p_2 \omega' \text{tr} \Delta \Sigma^2 T(\theta)^{-2} + p_2 u \text{tr} \Delta \Sigma T(\theta; p_1)^2 T(\theta)^{-2}.\end{aligned}\tag{15}$$

Special Cases of Theorem 2 In the limit $p_2 \rightarrow 0^+$ (i.e., no synthetic data; all the training data is real), $\zeta \rightarrow 0$ in Theorem 2, and we recover the main result of Bach (2023) as a special case, namely $E_{test}(\hat{f}_{RP}) \simeq B + V$, with B and V as given in the theorem. Note that even in this special case, our result is more general since it covers the entire regularization path while the formulae in Bach (2023) are only for the unregularized case $\lambda \rightarrow 0^+$. On the other hand, Theorem 2 is a generalization of Theorem 1, as can be seen by taking $\psi \rightarrow \infty$. Refer to Appendix F.3 for details.

3.3 Some Consequences of Theorem 2

We now explore a few important consequences of Theorem 2.

A Double-Descent Curve. The bias-variance decomposition presented in Theorem 2 is empirically illustrated in Figures 2 and 5 for the Gaussian setting (1) (see Appendix A.1 for details on the experimental setup). Notice the perfect match with experiments. The shape of the bias curve in Figure 2 (leftmost plot) is reminiscent of the well-known double-descent (Bach, 2023) in the unregularized setting $\lambda \rightarrow 0^+$. The divergence at the interpolation threshold $m = n$ (i.e. $\psi = 1$) is because the bias term B , the variance term V , and the extra term ζ (responsible for model collapse) all diverge to infinity at this point.

Strong Model Collapse. Observe that the first term in the expression for ζ given in Theorem 2 is lower-bounded by $p_2^2 \text{tr} \Delta \Sigma^3 (\Sigma + \theta I_d)^{-2}$, which scales linearly with the square of the proportion $p_2 \simeq n_2/n$ of synthetic data in the training dataset $\mathcal{D}$. However, unless $p_2 \rightarrow 0^+$, i.e. unless the proportion p_2 of synthetic data in the training dataset vanishes, the performance of the model eventually plateaus above the baseline $\bar{E}$ (corresponding to the setting where all training data is real, i.e. no synthetic data in training dataset). This is strong model collapse.

Since the factor $\text{tr} \Delta \Sigma_3 (\Sigma + \theta I_d)^{-2}$ only depends on the design choices of the model (via the scalar θ defined previously), we expect that different design choices (e.g., model size) will lead to different model collapse profiles. Figure 4 shows results for different proportions of synthetic data and various values of the quality parameters c^2 (cf. Definition 1), and also different model sizes m , in the toy setting of multivariate Gaussians with random projections (experimental details in Section A.1).

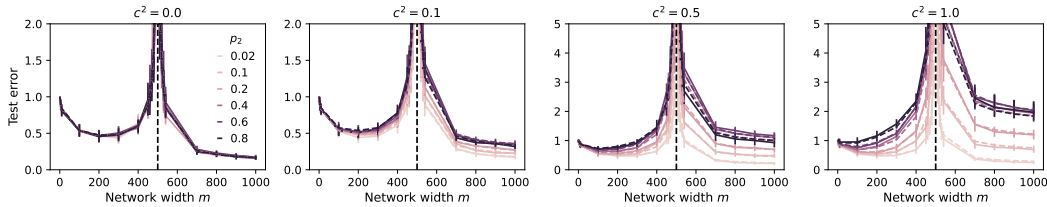


Figure 4: **Impact of model size (network width m) on model collapse.** For various levels of data quality c^2 (cf. Definition 1) we show test error as a function of model size for various mixing ratios (darker curves correspond to higher fractions p_2 of synthetic data). Error bars correspond to 5 independent runs.

Are Larger Models More Prone or Less Prone to Model Collapse? Figure 1 shows the results of a small experiment to investigate this. The input dimension is $d = 600$, and the covariance matrix is identity I_d (isotropic features). The total number of training examples is fixed to $n = 500$. The Δ matrix is taken to be of the form $\Delta = (c^2/d)\Sigma^{-1}$ (similar results are observed for different covariance matrices) for different values of c^2 as follows: $c^2 = 0$ (synthetic data of very high quality), represented with square markers; $c^2 = 0.1$ (high quality synthetic data), represented with

diamonds; $c^2 = 0.5$ (low quality), represented by triangles; and $c^2 = 1$ (very low-quality synthetic data), represented by crosses. As indicated on the figure, the leftmost plot corresponds to the regime where there is much fewer synthetic than real samples ($n_2 = 50$ synthetic samples versus $n_1 = 450$ real samples). Here, for both high-quality and low-quality synthetic data (squares), the optimal tradeoff is struck by larger models (i.e, larger values of ψ). For lower-quality data, the frontier shifts upwards and from left to right; **larger models cease being optimal for coping with model collapse and the optimal model size is intermediate.**

In the middle plot of Figure 1, size of the synthetic dataset is comparable to the size of the real dataset ($n_2 = 200$ versus $n_1 = 300$). For high-quality synthetic data, larger models are still better than smaller models. However, for this setting, the frontier shifts upwards and from left to right, and the optimal model size is intermediate. For the rightmost plot, the size of the synthetic dataset is considerably larger than the real dataset ($n_2 = 400$ versus $n_1 = 100$). The results are similar to the case $n_2 = 200$ except that the Pareto frontiers are higher over the diagonal (indicating more serious model collapse). In all cases, very small models are never optimal: they are not good even in the classical sense when training is done only on real data, and the presence of synthetic data only makes this worse.

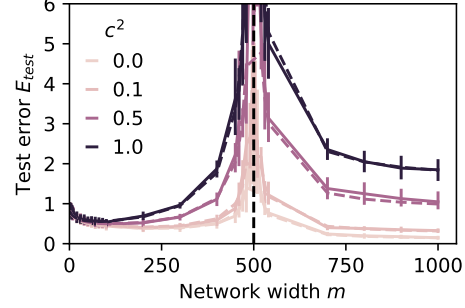


Figure 5: Impact of model size (network width m) on model collapse. As usual, solid curves correspond to experimental results (5 runs), while broken curves correspond to predictions of our theory (here, Corollary 4). Error bars correspond to 5 independent runs. Also see Figures 2 and 4.

4 Experimental Results

Our theoretical framework is developed within the context of high-dimensional linear regression and random projections models using Gaussian data. We now provide evidence that our theory is applicable to large-scale problems beyond Gaussian data: (1) neural networks on MNIST (Deng, 2012) and (2) language modelling.

4.1 Experiments on Neural Networks on MNIST

Our first departure from the confines of our theory are experiments with two-layer neural networks trained on the MNIST dataset (Deng, 2012) both in the random feature model (with ReLU activations) and with fully trained networks. These experiments complement the empirical validation we already provided in Figures 1, 2, 5, 4, and discussed amply at the end of Section 3.2.

Setup. For two-layer neural networks, we consider two scenarios: (1) learning with a random projection model as in Section 3.2, where the first layer of the network is fixed randomly, and only the second layer is trained, and (2) learning with a fully trainable neural network. The first setting directly corresponds to our theoretical results from Section 3.2, but with ReLU activation functions. In the case of fully trained neural networks in the second setting, our theory does not apply directly. However, we hypothesize that the general trends observed in our asymptotic theory will still hold: (1) there will be a significant model collapse, which only diminishes as the fraction of synthetic data approaches 0; (2) larger models will exhibit a more severe model collapse.

To align with the theoretical setting, we employ a (multivariate) regression approach where labels are converted to one-hot vectors and the model is trained using mean squared error. The synthetic labels were generated by another two-layer network, with Gaussian label noise (standard deviation of 0.1) added. A validation set is used to select the

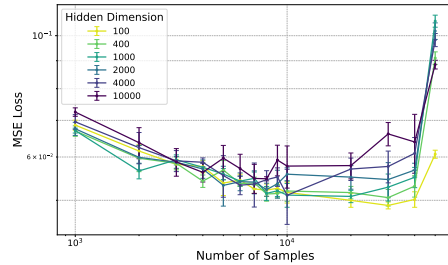


Figure 6: Fully trained two-layer network on MNIST data. Impact of model size (hidden dimension, aka network width) on model collapse. Here, the model is trained solely on synthetic data (i.e $p_2 \rightarrow 1$).